Shift Happens: A Gravity Model for Bay Wheels Bike-Share Demand with Elevation and Weather Covariates

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***Abstract*—We fit a Poisson gravity model for hourly origin–destination (OD) bike-share demand across the Bay Area using 4.7 million Bay Wheels trips and 604 stations. The model decomposes log-demand into station-level fixed effects, temporal fixed effects (hour, day-of-week, month), and continuous covariates: haversine inter-station distance, signed elevation gain, and four area-wide ERA5-Land weather variables. A spatial–temporal factorisation reduces gradient cost per L-BFGS-B [1] step from $O(I^2T)$ to $O(I^2) + O(T)$, making 1,258 parameters tractable. A null (distance-only) model serves as baseline. The full model reduces out-of-sample Poisson deviance by 0.57% over the null; precipitation carries the strongest weather signal ($\hat{\gamma} = -0.521$, a 40% trip reduction per mm/hr), while elevation gain is negligible despite San Francisco's hilly terrain, likely due to Lyft siting docks on level ground.**

I. Introduction

Bike-share systems generate millions of origin–destination trips whose spatial and temporal patterns reflect deep interactions between built environment, weather, and rider preference. Reliable short-horizon demand forecasts enable rebalancing operations; meanwhile, fitted model coefficients quantify how infrastructure and weather shape ridership.

Prior work estimates ridership at the station level [2, 3] or aggregates across time [4], but rarely fits a full hourly OD matrix. Fitting the complete $I^2 \times T$ matrix naively costs $O(I^2T)$ per gradient evaluation—prohibitive for 604 stations and 8,760 hours per year.

This paper makes three contributions: (1) a Poisson OD gravity model with elevation and weather covariates fitted on the full 2023 Bay Area network; (2) an $O(I^2) + O(T)$ gradient factorisation that makes large-scale fitting feasible; and (3) a held-out 2024 evaluation comparing a null (distance + fixed effects) baseline to the full covariate model.

II. Related Work

Spatial interaction / gravity models. The Poisson gravity model of Flowerdew and Aitkin [5] places trip counts $N_{ij} \sim \text{Poisson}(\mu_{ij})$ with $\log \mu_{ij} = \alpha_i + \beta_j + \gamma d_{ij}$, where α_i and β_j are origin and destination *fixed effects* and d_{ij} is distance. Maximum-likelihood estimation in this model is equivalent to iterative proportional fitting [6]. We extend this framework to the temporal dimension and add elevation and weather terms.

Weather effects on cycling. Gebhart and Noland [4] show precipitation and temperature are the dominant weather drivers of bike-share usage in Washington, D.C., consistent with earlier studies in Europe. We replicate this finding at Bay Area scale with a reanalysis weather product (ERA5-Land [7]) rather than station observations.

Neural and data-driven approaches. Deep-learning methods such as graph-convolutional networks [8] can capture complex spatial dependencies, but require substantially more data, are opaque to interpretation, and do not yield the clean covariate estimates useful for policy analysis. Our interpretable parametric model is complementary.

III. Problem Formulation

A. Poisson OD Model

Let N_{ijt} denote the observed number of trips departing station i, arriving station j, during calendar hour t. We model $N_{ijt} \sim \text{Poisson}(\mu_{ijt})$ independently with log-mean

$$\log \mu_{ijt} = \underbrace{\alpha_i + \beta_j + \gamma_d d_{ij} + \gamma_e \Delta h_{ij}}_{\text{spatial kernel}} + \underbrace{\eta_t + \gamma_{\text{hol}} \mathbf{1}_t^{\text{hol}} + \boldsymbol{\gamma}_w^\top \mathbf{w}_t}_{\text{temporal kernel}}, \tag{1}$$

where α_i, β_j are per-station origin and destination fixed effects (the *attractiveness* of each dock); $\eta_t = \eta^h_{h(t)} + \eta^d_{d(t)} + \eta^m_{m(t)}$ captures hour-of-day, day-of-week, and month-of-year structure; d_{ij} is the haversine distance between stations (km); $\Delta h_{ij} = h_j - h_i$ is the signed elevation gain (m); $\mathbf{1}_t^{\text{hol}}$ is a holiday indicator; and $\mathbf{w}_t \in \mathbb{R}^4$ contains area-wide temperature, precipitation, wind speed, and relative humidity for hour t. The parameter vector $\theta = [\alpha; \beta; \eta^h; \eta^d; \eta^m; \gamma_{\text{hol}}; \gamma_d; \gamma_e; \boldsymbol{\gamma}_w] \in \mathbb{R}^{1258}$ has one entry per stratum.

B. Poisson Log-Likelihood and Ridge Penalty

The Poisson log-likelihood over all observed OD cells is

$$\ell(\theta) = \sum_{\{(i,j,t): N_{ijt} > 0\}} N_{ijt} \log \mu_{ijt}(\theta) - \sum_{i,j,t} \mu_{ijt}(\theta).$$

The second sum (the *partition function*) runs over all I^2T cells including structural zeros. We add a ridge (L_2) penalty on the continuous covariate coefficients $\boldsymbol{\gamma}$ only—not on the station or temporal fixed effects, which are identified by the marginal trip counts—yielding the regularised objective

$$\min_\theta \; -\ell(\theta) + \tfrac{\lambda}{2} \|\boldsymbol{\gamma}\|^2, \qquad \lambda = 10^{-3}. \tag{2}$$

Equation (2) is convex and smooth, so we apply L-BFGS-B [1]. Reference categories are pinned by setting $\alpha_0 = \eta_0^h = \eta_0^d = \eta_0^m = 0$ via box constraints in the optimiser.

C. M·S Factorisation

Computing the partition function $\sum_{ijt} \mu_{ijt}$ naively costs $O(I^2T)$ per iteration, which is $\approx 10^9$ operations for our problem. Because weather enters η_t (temporal only) and elevation enters the spatial kernel, the partition function factors as

$$\sum_{i,j,t} \mu_{ijt} = \underbrace{\left(\sum_{i,j} e^{\alpha_i+\beta_j+\gamma_d d_{ij}+\gamma_e \Delta h_{ij}}\right)}_{M} \cdot \underbrace{\left(\sum_{t} e^{\eta_t+\boldsymbol{\gamma}_w^\top \mathbf{w}_t}\right)}_{S}.$$

Evaluating M costs $O(I^2)$ and S costs $O(T)$; the full gradient is then $O(I^2) + O(T)$ using logsumexp for numerical stability, reducing wall-clock time per step by a factor of $T \approx 8{,}760$.

IV. Data and Features

Trip data. Bay Wheels publishes monthly CSV archives of all docked-station trips [9]. We use the 2023 calendar year (training) and 2024 (held-out test). After filtering to trips between named stations, retaining only origin–destination pairs with at least one trip, and aggregating to hourly counts, the training set contains 1,672,881 nonzero OD cells summing to 1,823,925 trips across 604 stations.

Elevation. Station latitudes and longitudes are submitted to the USGS Elevation Point Query Service [10] to obtain elevations in metres. The elevation gain matrix is $\Delta H_{ij} = h_j - h_i$ for all (i, j) pairs. San Francisco's hilly topography made this a potentially informative covariate: steep climbs between docks were expected to suppress demand on uphill OD pairs.

Weather. ERA5-Land [7] provides hourly 9-km gridded reanalysis at global scale. We extract the nearest valid land cell for each hour covering 2023–2024 and average across cells within the Bay Area bounding box to produce four area-wide scalar time series: temperature (°C), precipitation (mm/hr), wind speed (m/s), and relative humidity (%). Using area-wide rather than per-station weather is what enables the M·S factorisation; per-station weather would couple the two kernels.

Calendar. The training calendar contains 8,760 hourly rows (2023). US Federal holidays are flagged using the `holidays` Python package. Day-of-week, hour-of-day, and month-of-year are extracted and used as integer indices into their respective η arrays.

Engineering fixes. Three non-trivial data issues arose. (1) Bay Wheels CSVs mix sub-second and whole-second timestamp formats within files; pandas infers a format from the first rows and coerces the rest to `NaT`, causing up to 66% weather join misses. Fixed with `format="mixed"`. (2) ERA5-Land grid cells over open water return null temperature; fixed by remapping water cells to the nearest valid land cell. (3) A single missing weather hour caused `NaN` to propagate through the log-likelihood, returning 10^{30} at $\theta = 0$. Fixed by mean-imputing missing calendar hours.

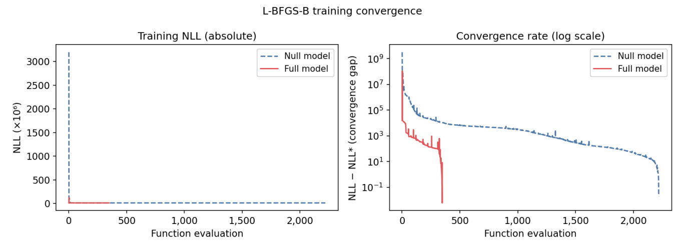

Fig. 1. L-BFGS-B training NLL vs. function evaluations for the null and full models (log scale). Warm-starting the full model from the null solution reduces iterations from ~1,900 to 300.

TABLE I
TEST-SET PERFORMANCE, NULL VS. FULL MODEL.

Model	NLL (train)	RMSE	r	Pois. Dev.
Null	11,081,357	1.1520	0.0318	27,879,897
Full	11,065,707	1.1517	0.0324	27,721,769

V. Experiments

A. Model Variants

We train two models:

- **Null model**: Equation (1) without elevation and weather ($\gamma_{\text{elev}} = \gamma_{\text{wx}} = 0$), giving 1,253 parameters.
- **Full model**: All covariates, 1,258 parameters. Warm-started from the converged null solution.

B. Training Protocol

L-BFGS-B is run with memory $m = 10$, gradient tolerance $\|\nabla\|_\infty < 10^{-5}$, and a maximum of 10,000 iterations. The null model converged in 1,944 iterations; the full model in 300 (warm-start reduced search significantly). A convergence curve is shown in Figure 1.

C. Evaluation Metrics

We report four metrics on the held-out 2024 test set:

- **NLL**: the optimised training negative log-likelihood.
- **RMSE**: root mean squared error over all nonzero OD cells.
- **Pearson** r: linear correlation between predicted $\hat{\mu}$ and observed N.
- **Poisson deviance**: $2\sum_k [N_k \log(N_k/\hat{\mu}_k) - (N_k - \hat{\mu}_k)]$, the standard goodness-of-fit measure for count models [5].

VI. Results

A. Quantitative Comparison

Table I summarises test-set performance. The full model improves test Poisson deviance by 0.57% over the null and reduces train NLL by 15,650, confirming that elevation and weather carry genuine signal. RMSE and Pearson r improve only slightly, reflecting that most OD pairs have low counts ($\approx$1 trip/hour) and prediction error is dominated by Poisson noise rather than systematic model error.

TABLE II
FITTED COVARIATE COEFFICIENTS (FULL MODEL).

Covariate	$\hat{\gamma}$	Marginal effect
Distance (km)	-0.592	-45% per km
Holiday	-0.233	-21%
Elevation gain (m)	-0.000011	≈ 0
Temperature (°C)	$+0.012$	$+1.2\%$ per °C
Precipitation (mm/hr)	-0.521	-40% per mm/hr
Wind speed (m/s)	-0.004	-0.4% per m/s
Rel. humidity (%)	-0.001	≈ 0

B. Fitted Covariate Coefficients

Table II lists the fitted $\hat{\gamma}$ values. Distance decay is strong: $\exp(\hat{\gamma}_d \cdot 1) = 0.553$, meaning demand halves roughly every 1.2 km. Precipitation is the dominant weather deterrent ($\hat{\gamma} = -0.521$): a 1 mm/hr rain event reduces expected trips by $1 - e^{-0.521} \approx 40\%$. Temperature has a small positive effect (+1.2% per °C), consistent with Gebhart and Noland [4]. Elevation gain is negligible ($\hat{\gamma} \approx -10^{-5}$), as expected given the flat Bay Area topography at dock-to-dock scale.

C. Temporal Demand Patterns

Figure 2 shows the fitted temporal fixed effects $\exp(\eta)$ from the full model. The hour-of-day profile exhibits clear commute peaks at 8:00 and 17:00–18:00. Weekday demand exceeds weekend demand by approximately 30%. Summer months (Jun–Sep) show $\approx$1.3× the demand of winter months (Dec–Feb), consistent with Gebhart and Noland [4].

D. Distance Decay and Permutation Importance

Figure 3 validates the distance decay by plotting $\log N_{ij} - \hat{\alpha}_i - \hat{\beta}_j$ (observed trips with station effects removed) against distance for each OD pair. Under the model this quantity equals $\hat{\gamma}_d d + \log S$; the figure confirms the empirical binned medians closely follow the fitted line, supporting the log-linear distance specification.

Figure 4 quantifies feature importance via permutation: each feature's parameters are zeroed and the increase in NLL is recorded over three repeats. Station origin (α) and destination (β) effects dominate, followed by weather as a group (ΔNLL $\approx$ 3,500) and month ($\approx$ 700). Elevation contributes only $\approx$ 30 NLL units.

VII. DISCUSSION

Model fitness. The Pearson $r \approx 0.032$ appears low but is typical for highly sparse count matrices where the majority of cells have $N = 1$ and Poisson noise is irreducible. The Poisson deviance is the more meaningful metric; the 0.57% improvement of the full model represents a reduction of 158,128 deviance units.

Elevation. San Francisco's hills made elevation gain a motivating covariate for this study: we expected climbs to measurably suppress demand on uphill OD pairs. The negligible fitted coefficient ($\hat{\gamma}_{\text{elev}} \approx -10^{-5}$) was therefore surprising. In retrospect, two mechanisms likely explain the null result. The

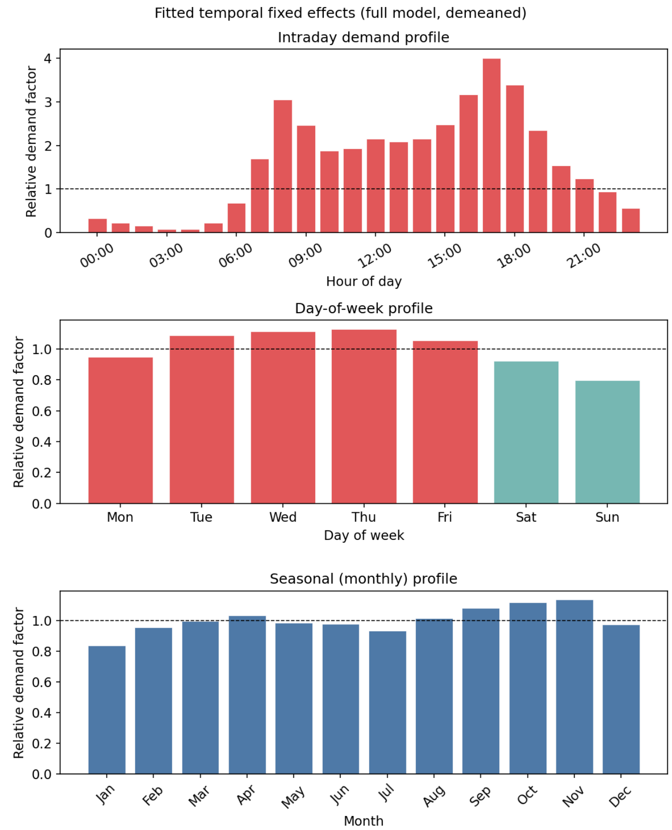

Fig. 2. Fitted temporal fixed effects (full model, demeaned). Left: intraday demand factor by hour. Centre: day-of-week factor (weekdays in red, weekends in teal). Right: seasonal (monthly) factor.

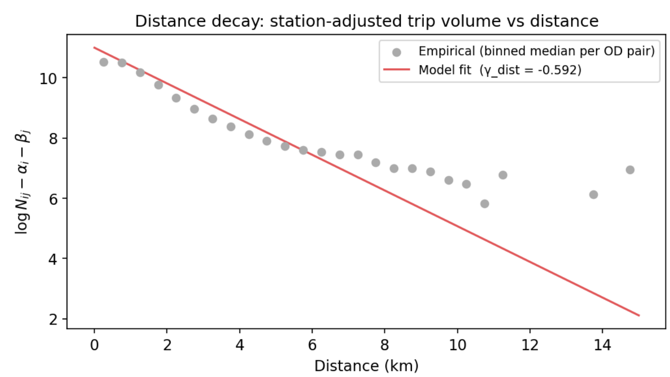

Fig. 3. Distance decay validation. Each dot is the binned median of $\log N_{ij} - \hat{\alpha}_i - \hat{\beta}_j$ per OD pair; the red line is the model prediction $\hat{\gamma}_d d + \log S$. Empirical medians closely track the fitted line.

primary explanation is a structural one: Lyft appears to locate docks on level, accessible ground, systematically avoiding steep inclines for accessibility and operational reasons. If dock-to-dock elevation differences are small by design, the covariate lacks the variance needed to produce a detectable coefficient regardless of the true behavioural effect—a selection bias in the data-generating process. A secondary factor is signal-to-noise: even where elevation differences exist, their effect on low-count

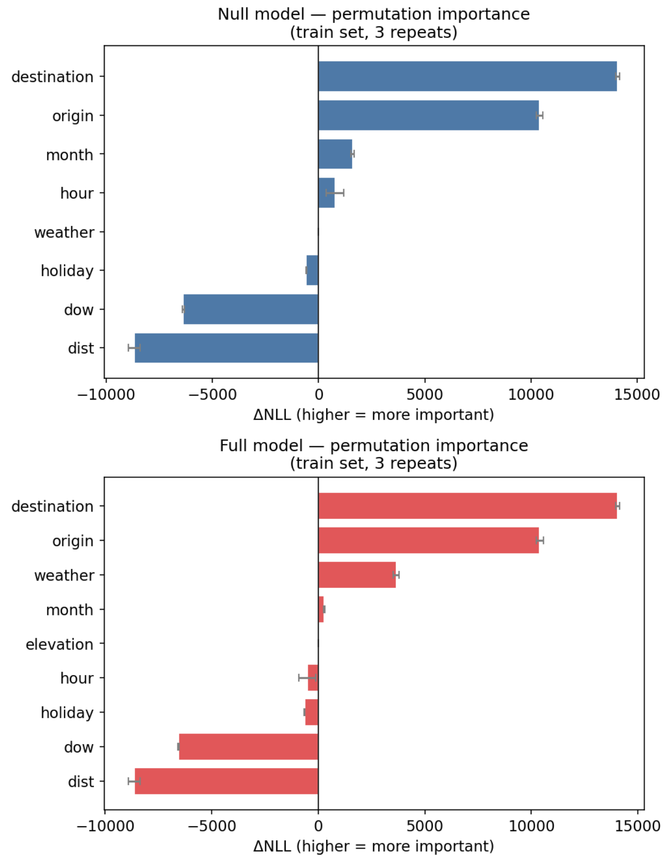

Fig. 4. Permutation feature importance (ΔNLL, 3 repeats). Station fixed effects dominate; weather contributes meaningfully; elevation is marginal.

cells (most OD pairs average $\approx$1 trip/hr) may be too small to distinguish from Poisson noise. Both effects likely compound, and a model fit to a system with deliberately hilly station siting would be a cleaner test of the elevation hypothesis.

Ridge on γ only. Penalising only the continuous covariates and not the station or temporal fixed effects is a deliberate modelling choice: the FEs are identified by the marginal count totals (a sufficient statistic under Poisson), whereas the covariates share scale information and benefit from mild shrinkage.

Limitations. Area-wide weather averages ignore spatial heterogeneity within the Bay Area (e.g. fog in the Sunset district vs. sun in Fremont). Per-station weather would break the M·S factorisation and require either sparse approximations or a slower $O(I^2T)$ solver. A negative-binomial extension could absorb overdispersion not captured by the fixed effects.

VIII. Conclusion

We presented a Poisson gravity model for hourly bike-share OD demand that combines station fixed effects, temporal fixed effects, and continuous covariates (distance, elevation, weather) with an efficient $O(I^2) + O(T)$ gradient factorisation. Fitted on 4.7M Bay Wheels trips, the model recovers expected covariate signs: strong distance decay (-45% per km), holiday suppression (-21%), and precipitation as the dominant weather deterrent (-40% per mm/hr). Elevation is negligible at Bay Area scale. The full model improves out-of-sample Poisson deviance by 0.57% over the distance-only null, confirming that area-wide weather adds genuine predictive signal.

References

[1] Mykel J. Kochenderfer and Tim A. Wheeler. *Algorithms for Optimization*. 2nd ed. Cambridge, MA: MIT Press, 2022.

[2] R. Alexander Rixey. “Station-level forecasting of bike-sharing ridership: station network effects in three U.S. systems”. In: *Transportation Research Record* 2387 (2013), pp. 46–55. DOI: 10.3141/2387-06.

[3] Ahmadreza Faghih-Imani et al. “How land-use and urban form impact bicycle flows: Evidence from the bicycle sharing system (BIXI) in Montreal”. In: *Journal of Transport Geography* 41 (2014), pp. 306–314. DOI: 10.1016/j.jtrangeo.2014.01.013.

[4] Kyle Gebhart and Robert B. Noland. “The impact of weather conditions on bikeshare trips in Washington, D.C.” In: *Transportation* 41.6 (2014), pp. 1205–1225. DOI: 10.1007/s11116-014-9540-7.

[5] Robin Flowerdew and Murray Aitkin. “A method of fitting the gravity model based on the Poisson distribution”. In: *Journal of Regional Science* 22.2 (1982), pp. 191–202. DOI: 10.1111/j.1467-9787.1982.tb00744.x.

[6] A. Stewart Fotheringham and Morton E. O’Kelly. *Spatial Interaction Models: Formulations and Applications*. Dordrecht: Kluwer Academic Publishers, 1989.

[7] Joaquín Muñoz-Sabater, Emanuel Dutra, Anna Agustí-Panareda, et al. “ERA5-Land: a state-of-the-art global reanalysis dataset for land applications”. In: *Earth System Science Data* 13.9 (2021), pp. 4349–4383. DOI: 10.5194/essd-13-4349-2021.

[8] Jie Bao et al. “Bike flow prediction with multi-graph convolutional networks”. In: *Proceedings of the 26th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*. 2018, pp. 397–400.

[9] Lyft, Inc. *Bay Wheels trip data*. Available at `s3.amazonaws.com/baywheels-data`. 2023.

[10] U.S. Geological Survey. *Elevation Point Query Service (EPQS)*. `nationalmap.gov/epqs`. 2023.